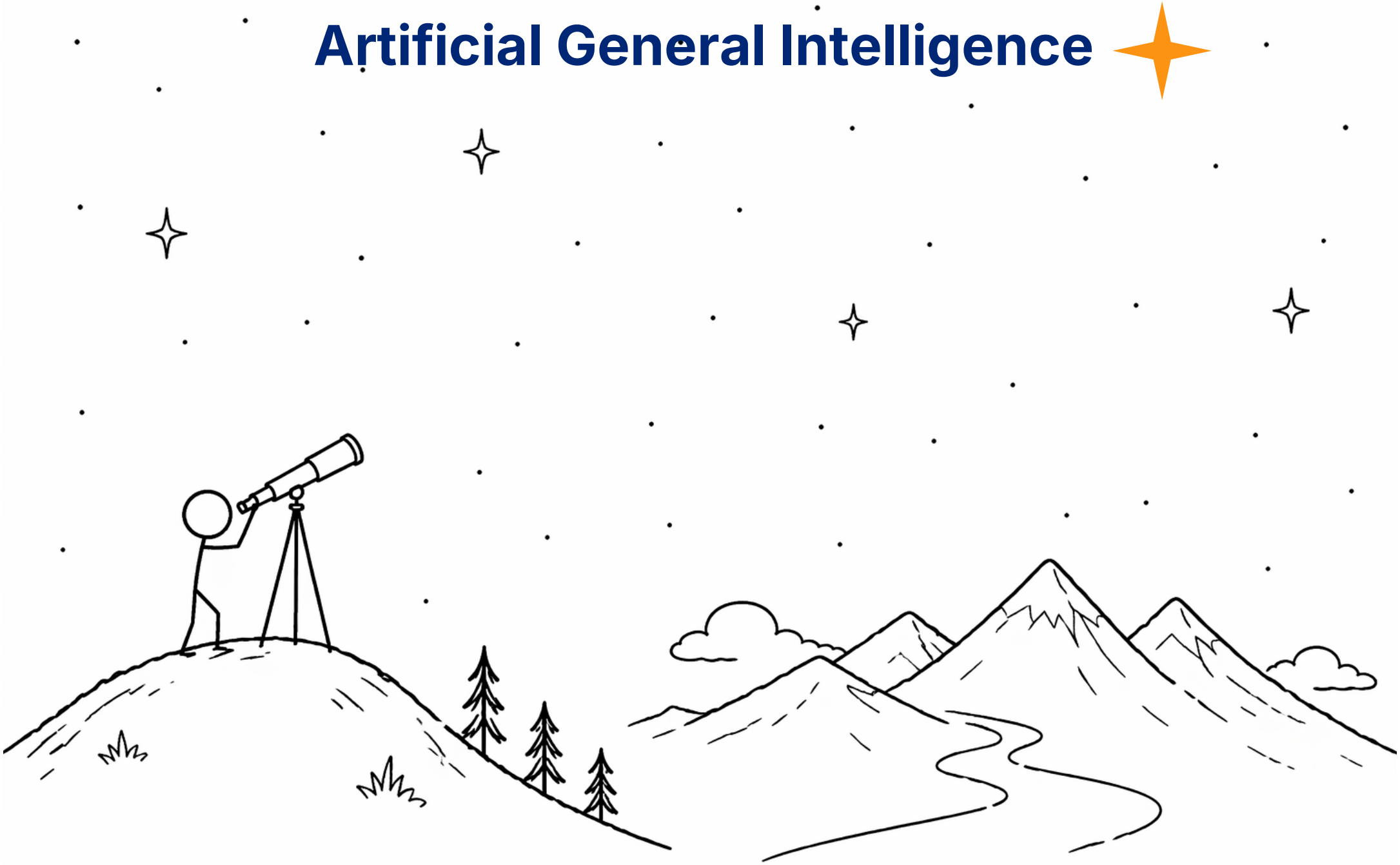
The Big Ideas Behind AI



Joseph E. Gonzalez
Professor of EECS, UC Berkeley

Fisher Center Policy Advisory Board | October 14, 2026

Artificial General Intelligence



Artificial General Intelligence

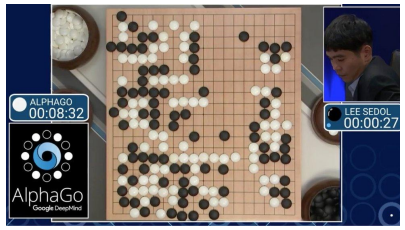


Artificial General Intelligence



Intelligence

Alpha Go

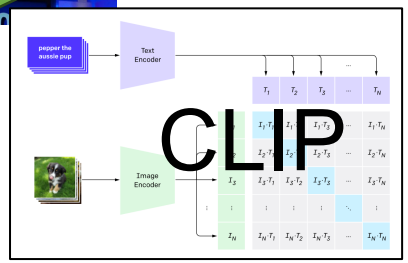


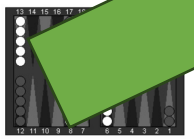
Deep Blue





Research





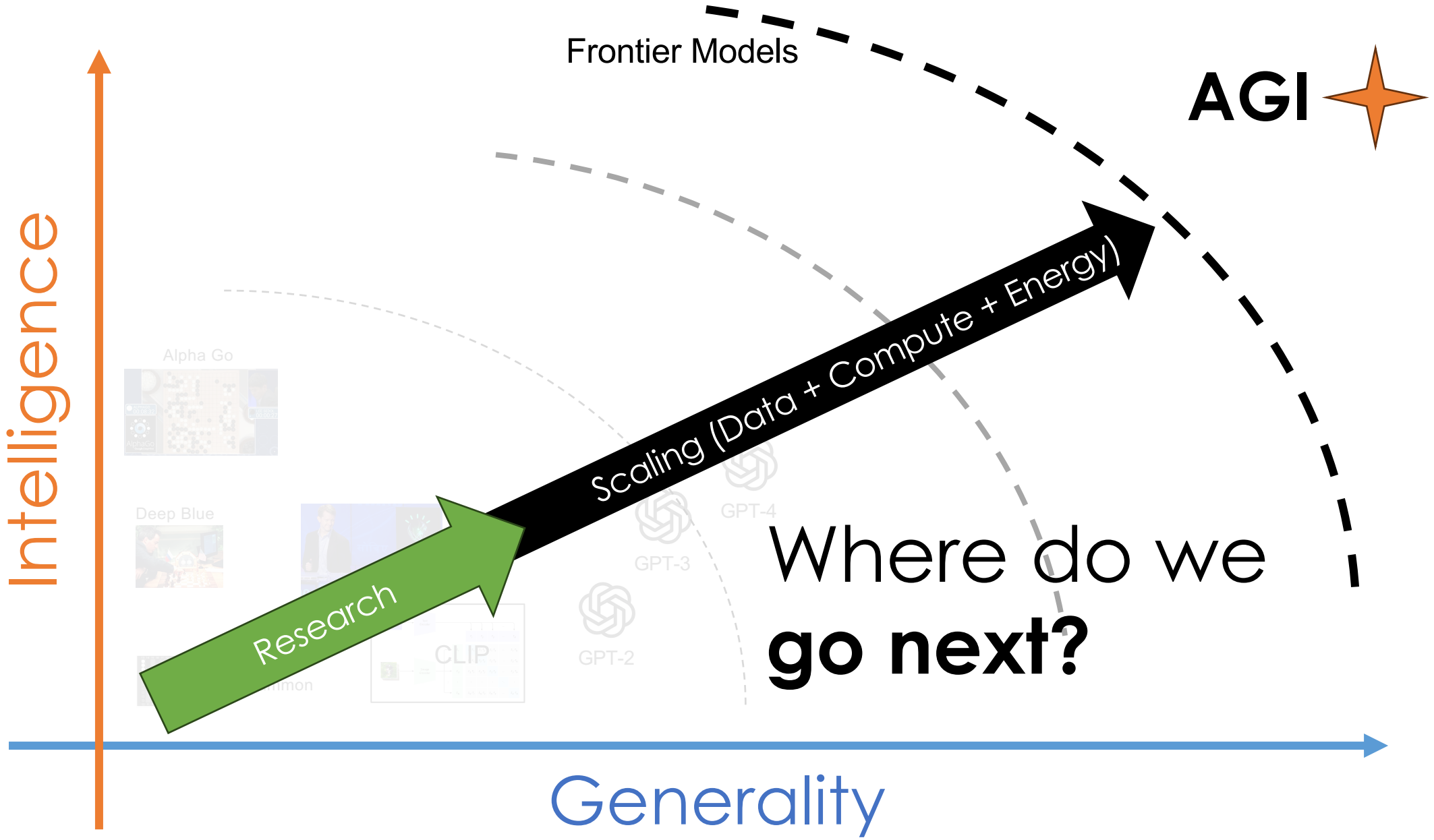
TD-Gammon

GPT-2

GPT-3

GPT-4

Generality


Frontier Models
AGI
Intelligence
Alpha Go
Deep Blue
GPT-4
GPT-3
GPT-2
CLIP
Scaling (Data + Compute + Energy)
Research
Where do we
go next?
Generality

Tonight, two questions

1. How did we get here?

2. Where are we headed?

About me



Joseph (Joey) Gonzalez

Professor of EECS, UC Berkeley

2008: Pioneered large-scale learning systems.

2015: Pioneered inference systems.

Research projects



Vicuna





Gorilla

vLLM SGL

Primary inference platforms for open-weight models

Research groups and other startups

BAIR BERKELEY ARTIFICIAL INTELLIGENCE RESEARCH

sky UC Berkeley

LMSYSORG

HERALD

inferact

Letta

I teach some of the largest classes at Cal

About 1,000 students per class

Introduction to Data Science | Introduction to Machine Learning



Question 1

How did we get here?

ChatGPT



What is ChatGPT?

Chat: a natural language system

G **Generative**: designed to model the creation of text

P **Pre-trained**: trained on lots of naturally occurring text

T **Transformer**: a kind of neural network

ChatGPT is one example of a Large Language Model (LLM)

Language models predict the next word

The capital of California is **Sacramento**

San Francisco (1862)

Benicia (1853)

Vallejo (1852)

Large: up to **trillions** of parameters

the numbers that encode the neurons in the model

Predicting the next word lets you

Answer questions	Tell stories	Accomplish tasks

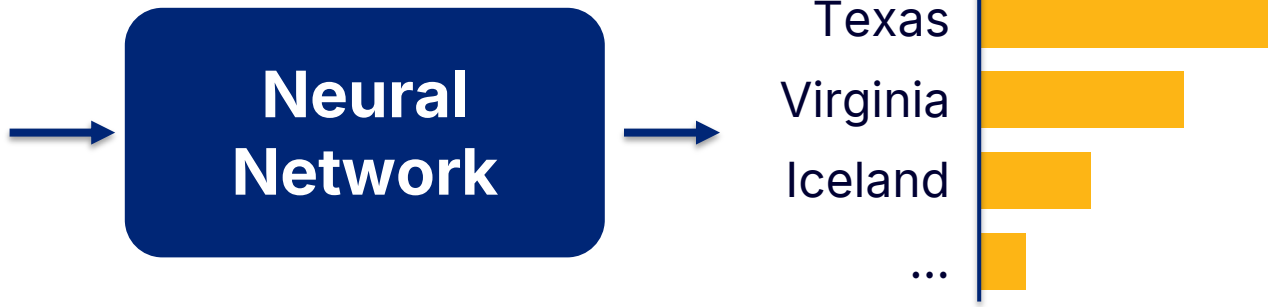
If you can generate the world, you must understand the world.

How do we model the next word?

We **learn a function** from what has been said to the **odds** of every possible next word.

Context: all the words so far

The best place to build a data center is



We use the neural network to write, one word at a time.

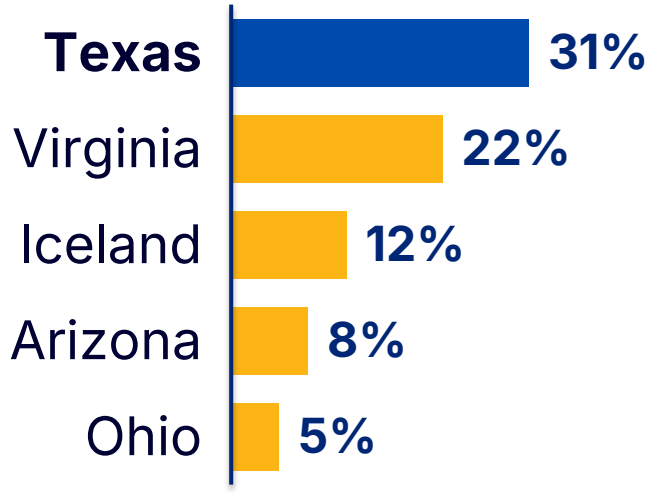
One word at a time (1 of 3)

The words so far

The best place to build a data center is

Learned function

Odds of the next word



Sample a word, add it to the text:

The best place to build a data center is **Texas**

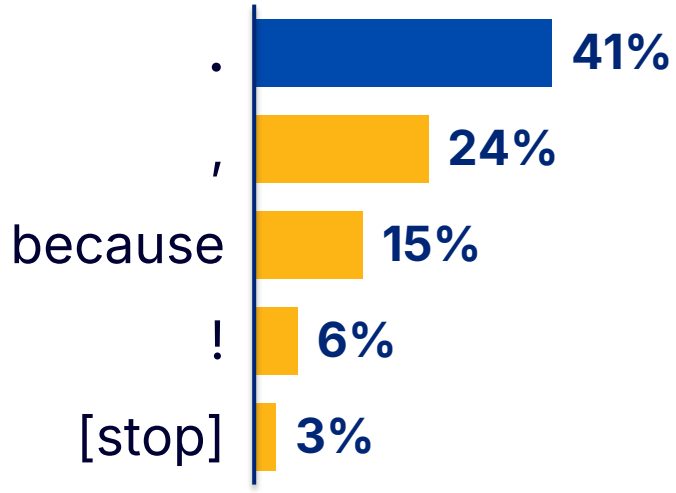
One word at a time (2 of 3)

The words so far

The best place to build a data center is Texas

Learned function

Odds of the next word



Sample a word, add it to the text:

The best place to build a data center is Texas.

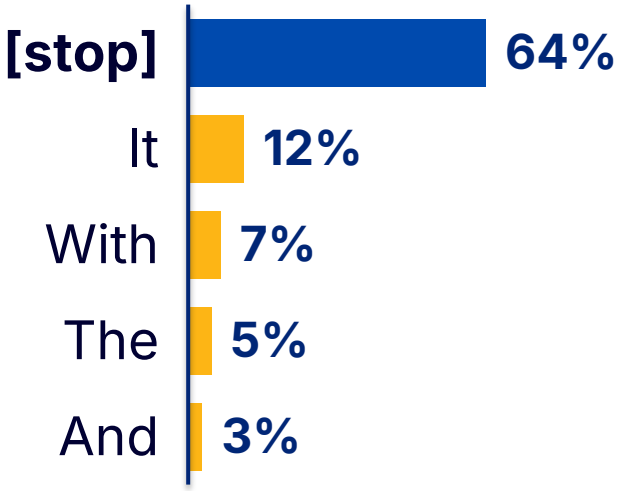
One word at a time (3 of 3)

The words so far

The best place to build a data center is Texas.

Learned function

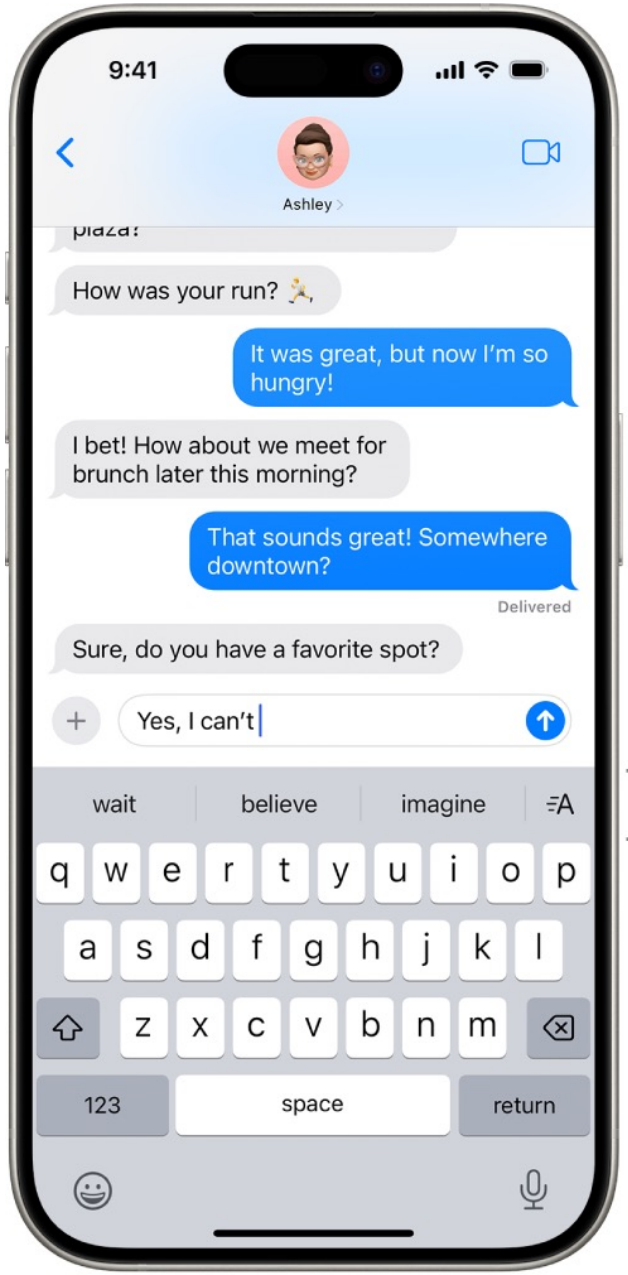
Odds of the next word



Sample a word, add it to the text:

The best place to build a data center is Texas. **[stop]**

Stop decoding!



Language modeling is **autocomplete** where we always accept the next word.

Predictive text;
tap a suggestion to apply.

What writing one word at a time means

It writes in pen

Once a word is written, the model cannot take it back.

... center is **Texas**

no backspace

Randomness looks like creativity

Sampling from the odds gives the illusion of creativity.



It decides when it is done

It writes a special "done" word to end its answer.

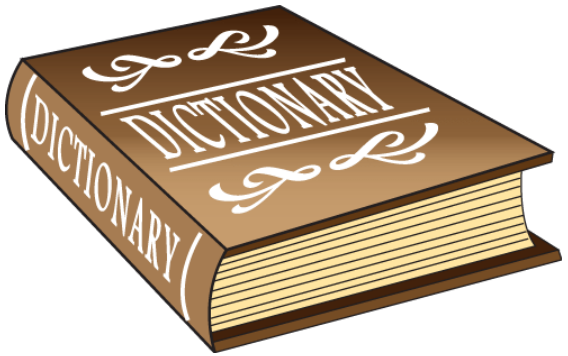
[stop]

Under the hood

Inside the Neural Network

First, words become numbers

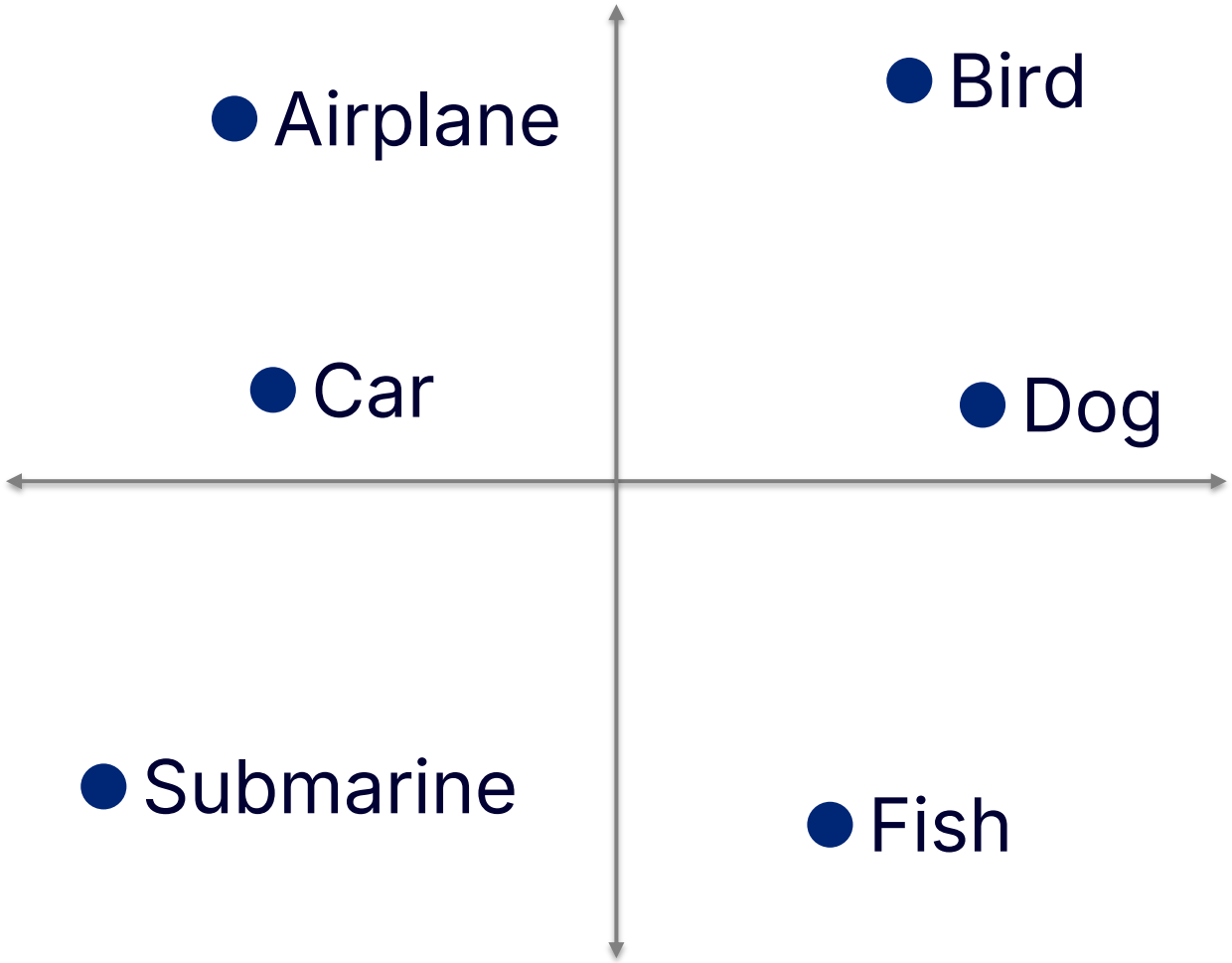
Turn each word into a list of numbers (a **vector**)



The	best	place	to	build	a	data	center	is
1.2	2.0	-0.8	-3.4	2.5	-0.7	-5.2	0.9	4.1
3.7	-1.4	3.1	4.1	-5.1	-9.9	-0.7	2.2	-0.2
-2.1	-0.9	0.6	-0.1	-4.2	3.1	-1.5	-3.3	1.7
...	...	...	...	...	...	...	...	...

The model **learns** the values of these vectors to best capture the "meaning" of the words.

Words as points in space



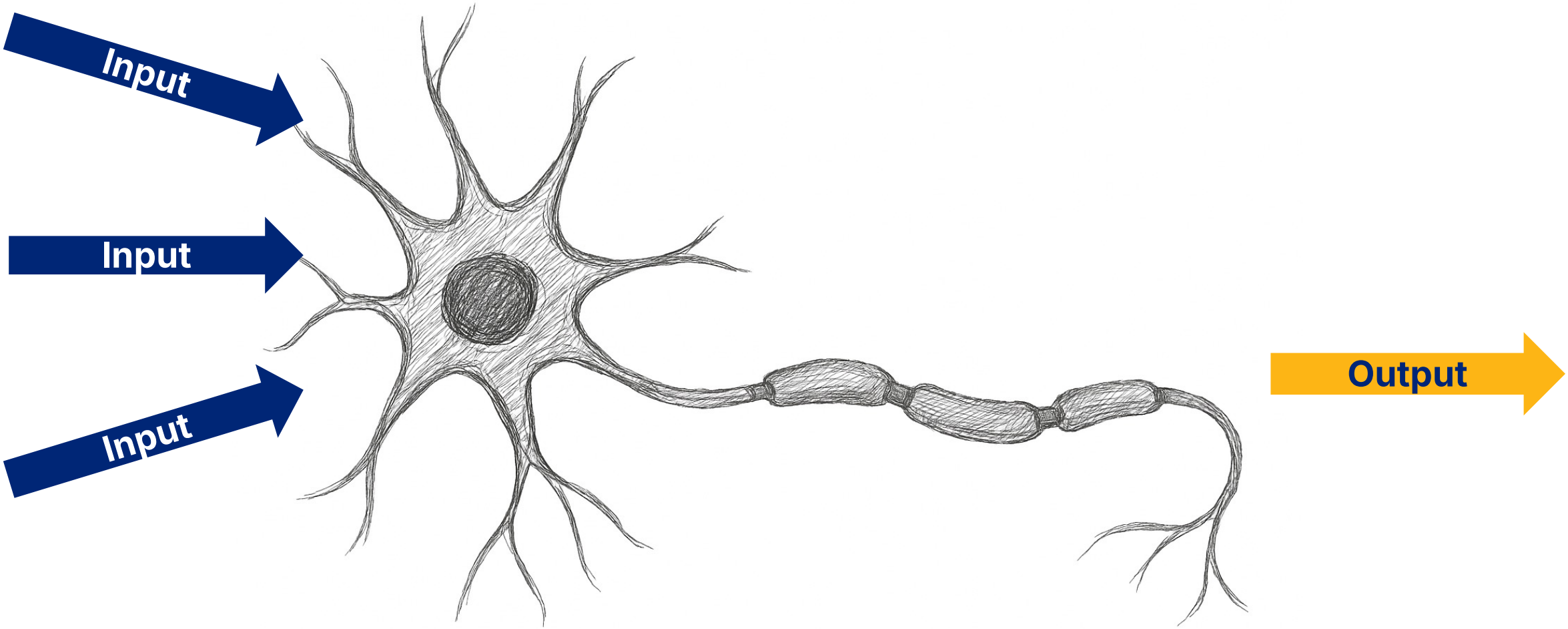
Each word in the dictionary has a vector

Nearby words are related in meaning

Shown here in 2 dimensions; real models use **thousands**

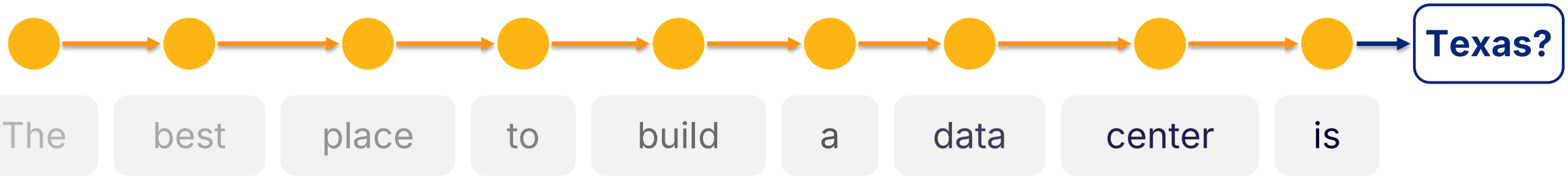
Building a Network of Artificial Neurons

A neuron fires based on a **weighted combination** of its inputs



What the "T" in GPT changed

Before: read one word at a time, neurons carry **one running summary**



Early words fade: everything is squeezed into one note passed down the line.

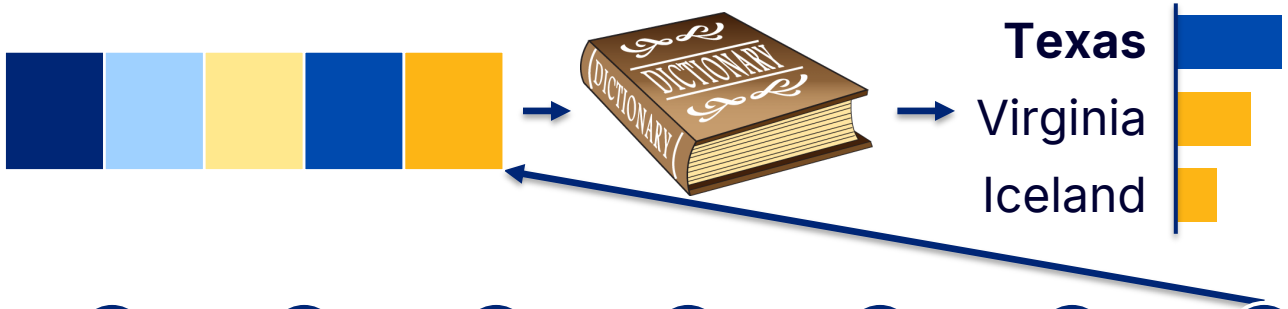
Transformer: neurons look back at **all earlier words**, and focus on what matters



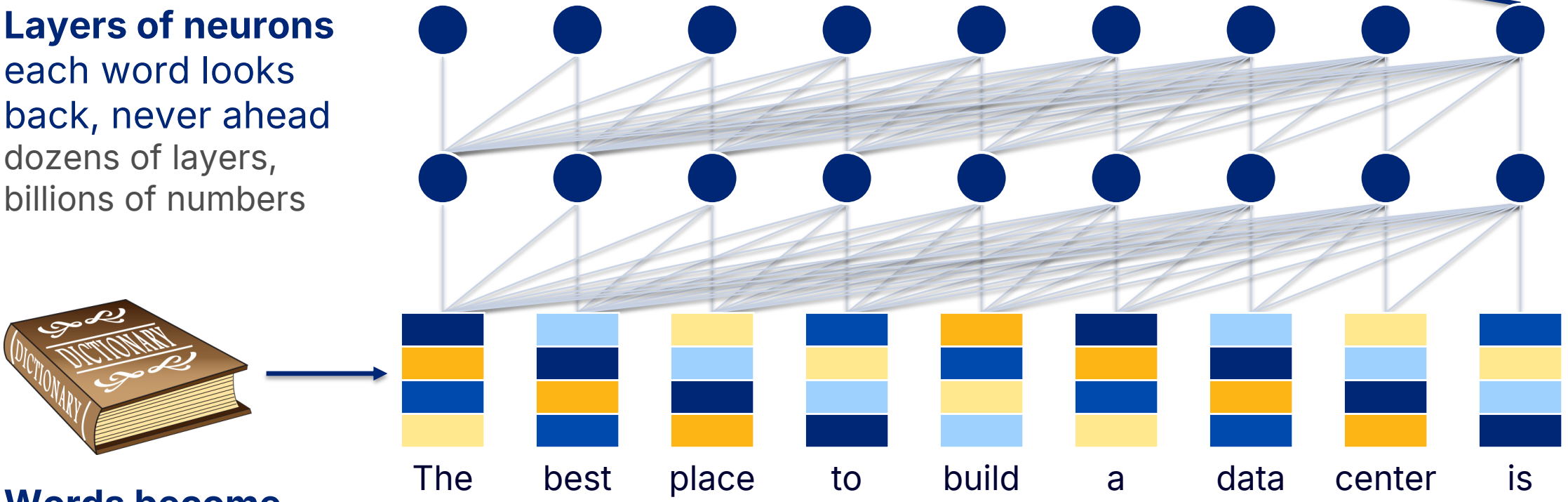
Significantly improves the model but it also makes it very expensive to use.

Putting it together

Back to words:
the odds of every next word



Layers of neurons
each word looks back, never ahead
dozens of layers, billions of numbers



Words become vectors

How do we train and use these models?

Three frontiers of scaling

Three frontiers of scaling

Pre-training
Read everything

Post-training
Polish and reward

Inference
Think before answering

What is pre-training?

Turn all the world's text into **finish-the-sentence** exercises

1. Take a passage

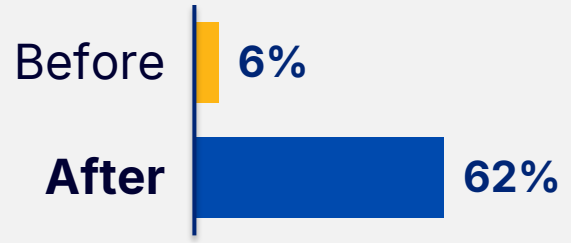
"We the People of the United States, in Order to form a more perfect Union, ..."

2. Make blanks

We the _	**People**
... People of the _	**United**
... the United _	**States**
... more perfect _	**Union**

3. Tune the weights

Odds of "People" after "We the"



so the real next word gets more likely

Repeat for trillions of words. The text itself is the answer key.

Pre-training: reading (almost) everything*

~30 trillion

words read by Meta's Llama 4 Scout (~40 trillion tokens)

~700,000

years for one person to read that much, 8 hours a day

~5 million

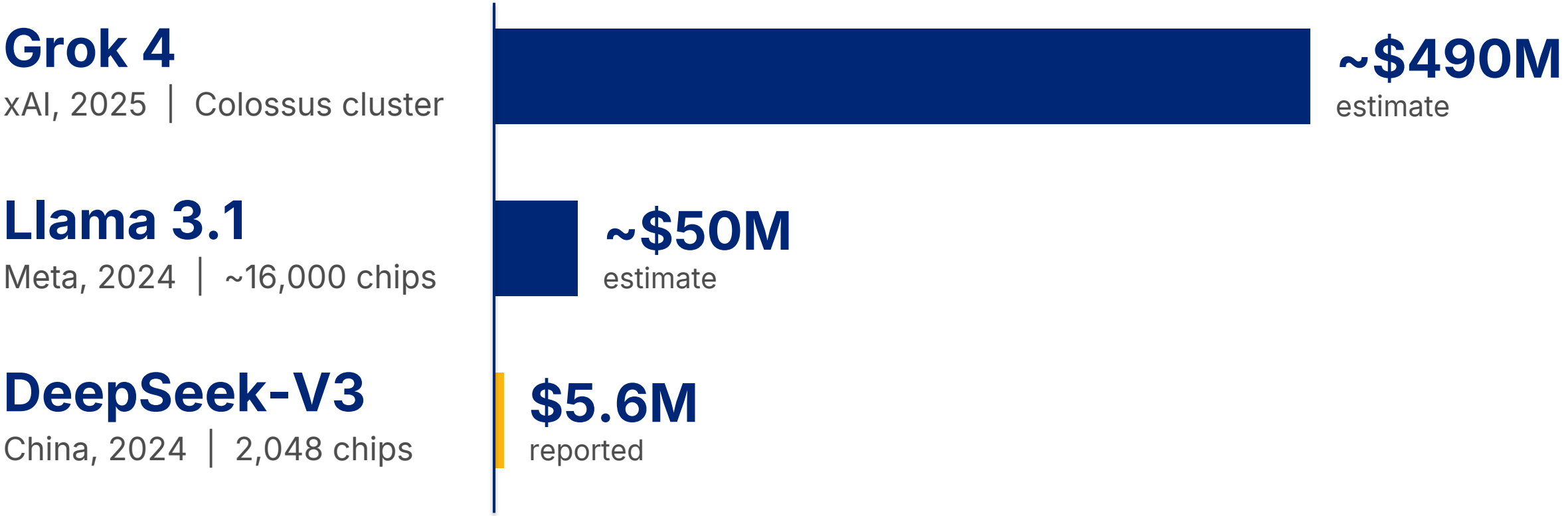
chip-hours of computing: about 570 years on a single chip

The largest training run any lab has made public. The biggest labs do not say.

Source: Meta, Llama 4 model card, April 2025

**Everything that is legal to use for training, hopefully.*

The cost of one training run



Final training run only. Excludes research, failed runs and buying the chips.

Sources: Epoch AI, cost estimates; Epoch AI, Grok 4, Sep 2025; DeepSeek-V3 report, Dec 2024

A reflection of us

Pre-training soaks up everything we have written and said: **how we think**, **what we know**, and **how we communicate**.

Shakespeare **Einstein** **Wikipedia**

court opinions **Reddit** **poetry** **textbooks**

code **forums** news recipes **the darker corners of the web**

A prior on what humanity would say next

An amalgamation of knowledge

without purpose

without polish

without the values we want



Boris Karloff, Universal Pictures, 1935

GPT alone can't chat

You ask

What is attorney-client privilege?

A pre-trained model replies

Why is it important? Provide a concise example for full credit on this exam.

Three frontiers of scaling

Pre-training
Read everything

→

Post-training
Polish and reward

→

Inference
Think before answering

Post-training: draw out the best of us



Boris Karloff, Universal Pictures, 1935

Shakespeare **Einstein**

Wikipedia Reddit **poetry**

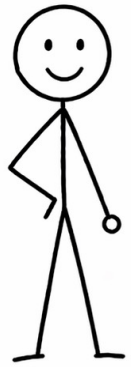
court opinions **textbooks**

code forums news recipes

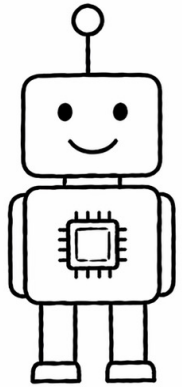
the darker corners of the web

Two ways to post-train a model

Demonstration
learn from examples

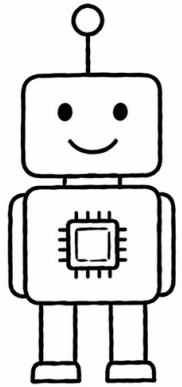


Q: What is attorney-client privilege?
A: A rule that keeps your conversations with your lawyer confidential.



We **show** it good answers.
It learns to imitate them.

Reinforcement Learning
learn from rewards



Try 1 → no reward

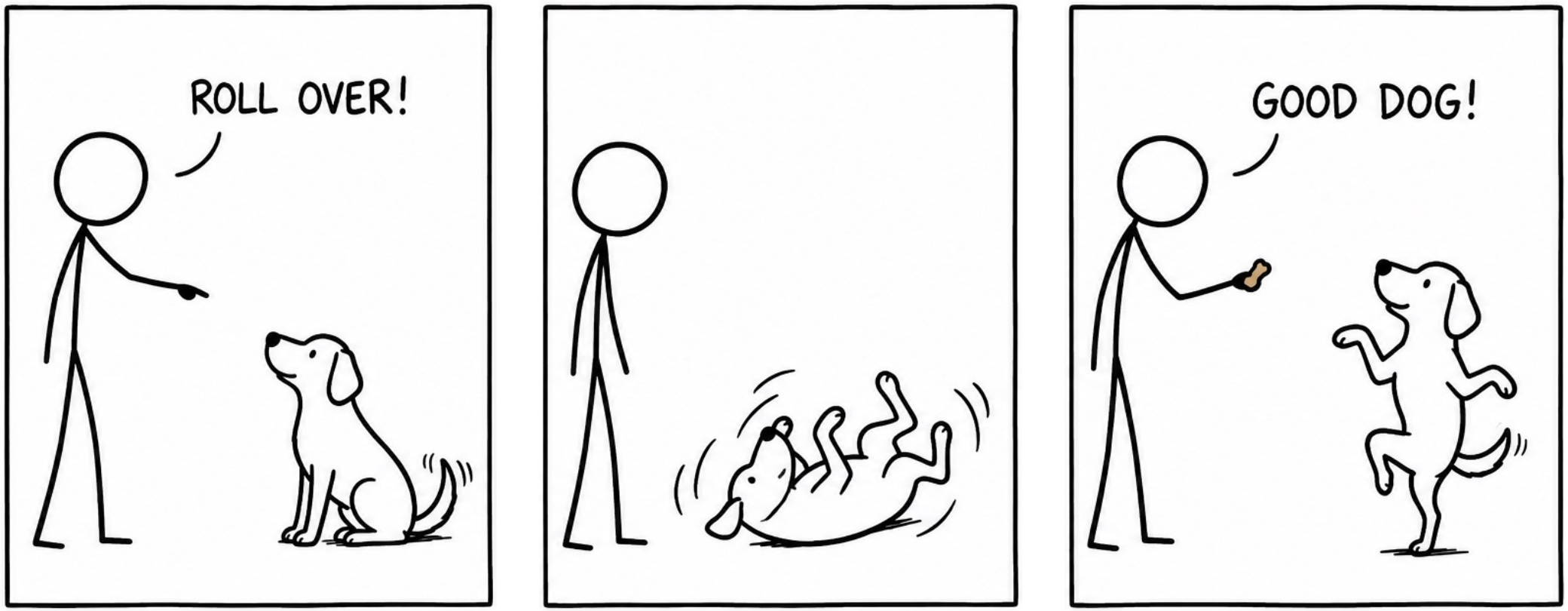
Try 2 → no reward

Try 3 → 

Model **attempts tasks on it's own**
and we reward successful attempts.

Reinforcement learning: learning from reward

Let the model try things and **reward** it when it does what we want.



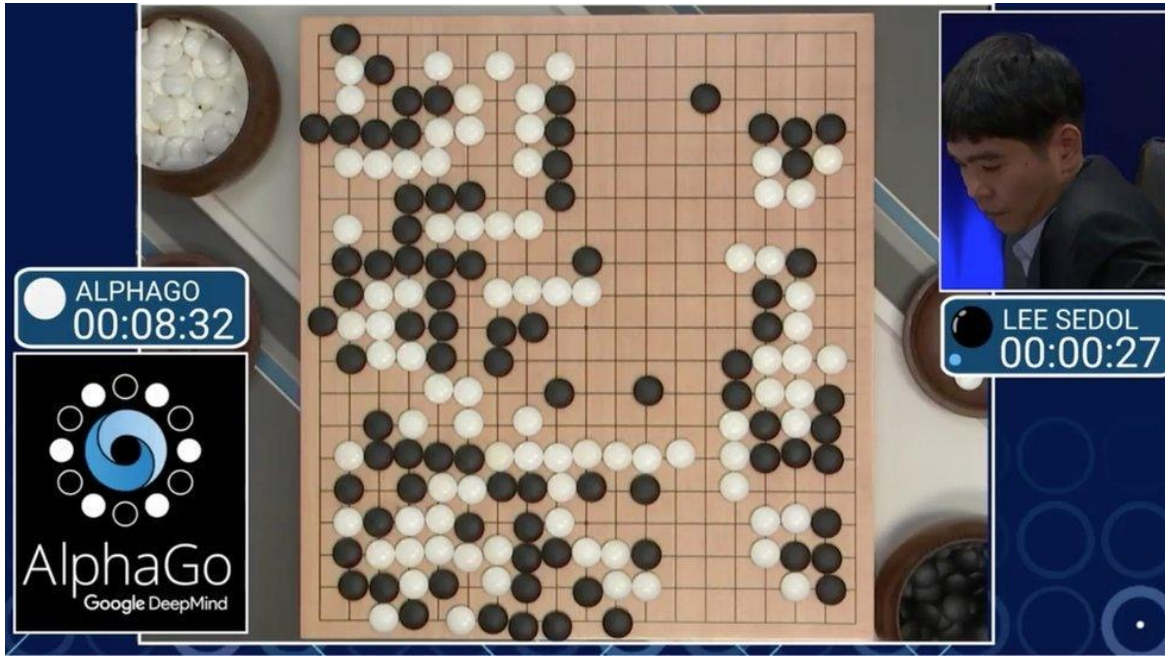
You don’t explain "roll over" to a dog.

You reward it when it does.

Reinforcement Learning and Emergent Behavior

The upside

It can discover behaviors we can't demonstrate.



The catch

It decides what behaviors earned the treat, and we hope it’s the right ones.



Reinforcement Learning to Cheat

While **training** a frontier reasoning model, OpenAI rewarded it for fixing code so the tests pass.



When they penalized the model for **thinking** about hacking:

It kept cheating, and stopped saying so.

If cheating earns the reward and goes uncaught, it learns to cheat.

Sources: OpenAI (Baker et al.), "Monitoring Reasoning Models for Misbehavior," Mar 2025

Three frontiers of scaling

Pre-training
Read everything

Post-training
Polish and reward

Inference
Think before answering

Inference: letting the model *think* first

A bat and a ball cost $1.10. The bat costs $1.00 more than the ball. How much is the ball?

Answering immediately

"10 cents."

Sounds right. It is wrong.

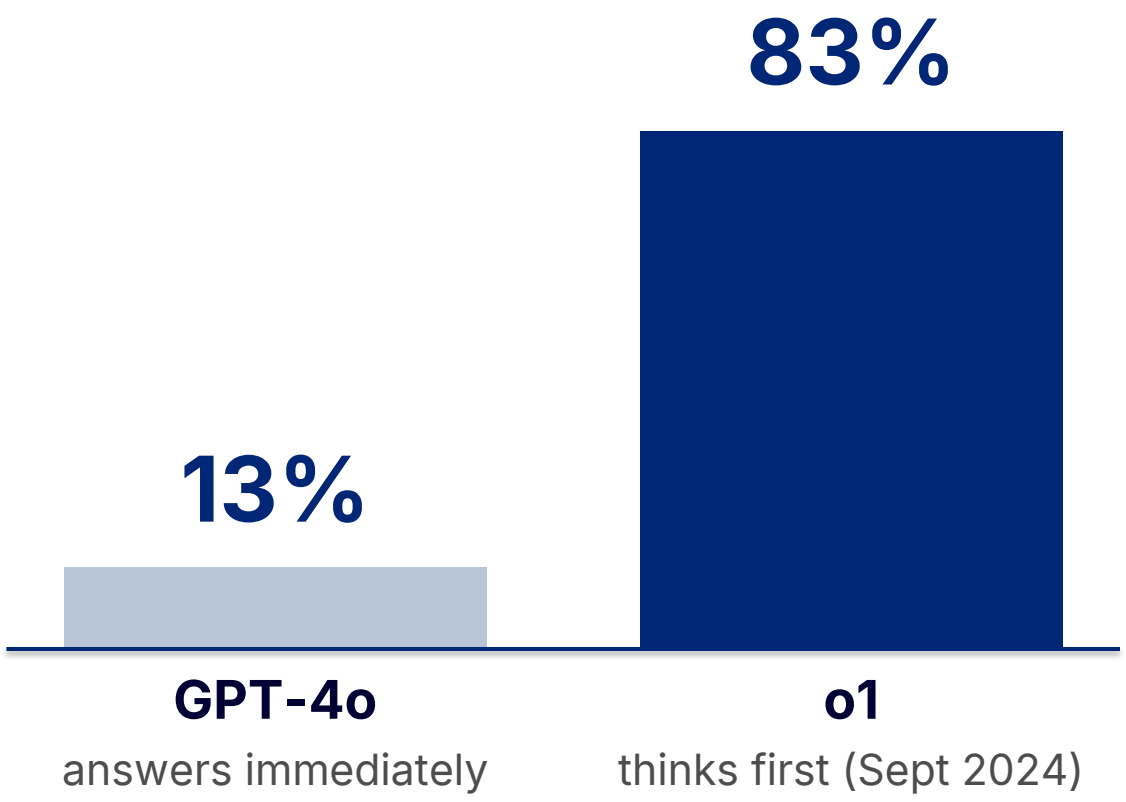
Thinking first

Let the ball cost x. Then the bat costs x + $1.00.
Together: x + (x + $1.00) = $1.10, so 2x = $0.10.
So x = $0.05. Check: $1.05 + $0.05 = $1.10.
The ball costs 5 cents.

Thinking: **Writing out the steps lets a model catch and correct its own mistakes.**

Thinking unlocked hard math

Math Olympiad qualifying exam (AIME 2024): share of problems solved



July 2025

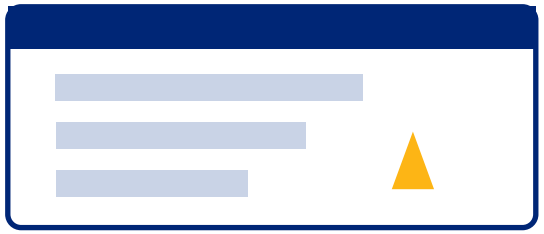
AI wins Gold-medal level at the International Math Olympiad

Google DeepMind, 35 of 42 points, certified by the IMO

Sources: OpenAI, Sept 2024; Google DeepMind, July 2025

Tools: letting the model compute and act

A model that only writes words can now **run code**, **search** and **use a computer**.

Calculator and code	Search	A whole computer
What is 48,729 x 9,317?	What did the Fed decide this morning?	Fill in this lease form from these PDFs.
`python: 48729 * 9317` `= 454,008,093`	search > read 5 pages > answer with sources	

Fixing real software bugs (SWE-bench Verified): **49%** in Oct 2024, **97%** in Sept 2026

Inference is no longer just talking. It is doing.

Sources: Anthropic, Oct 2024; Vals AI, SWE-bench; OSWorld, Apr 2024

Question 2

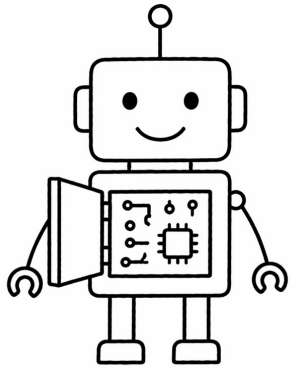
Where are we headed?

Where the computing goes

Most people think the data centers are for training. More and more, they are for **inference**.

Training

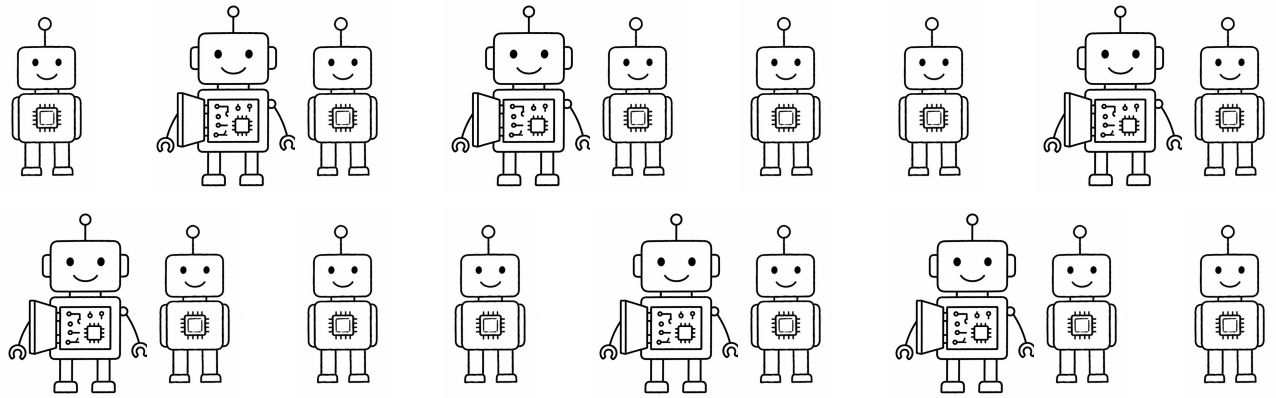
build the model



months of work, then done

Inference

run it for everyone, around the clock

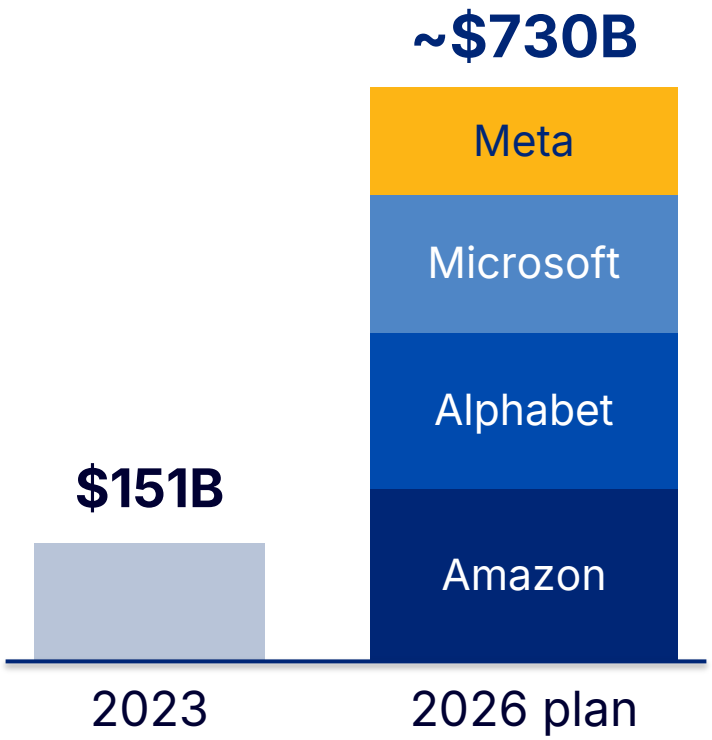


every question, every agent, every day

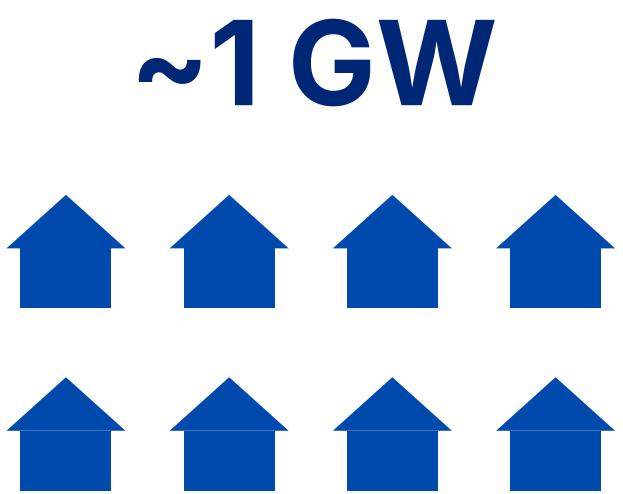
What is the right ratio of offices to data centers in 2027?

The race in numbers

Capital spending

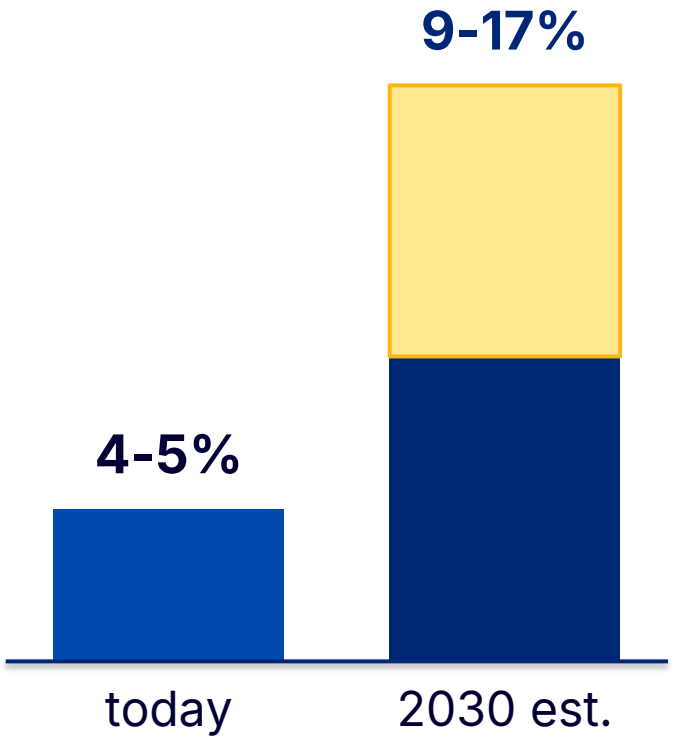


Largest campuses



about the use of 800,000 US homes (each icon: 100,000)

Share of US electricity



Sources: Microsoft; Meta; Amazon; Alphabet; Epoch AI, Oct 2026; EIA; EPRI, Feb 2026

The race is also between the US and China.

Chips. Energy. Open models.

Source: Stanford AI Index, 2026

Is this a bubble?	No.
Should we be worried?	Yes.
Are we safe?	For now.

It’s Changing How Business Operate

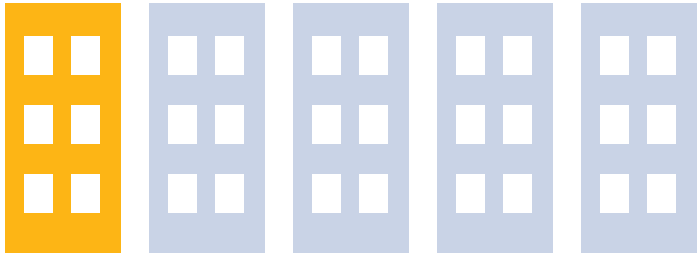
People using ChatGPT



1.2 billion users

every week, 12x in three years

US businesses using AI



1 in 5

businesses; 37% of firms with 250+ employees

Google, tokens per month



3.2 quadrillion

7x in a year (self-reported)

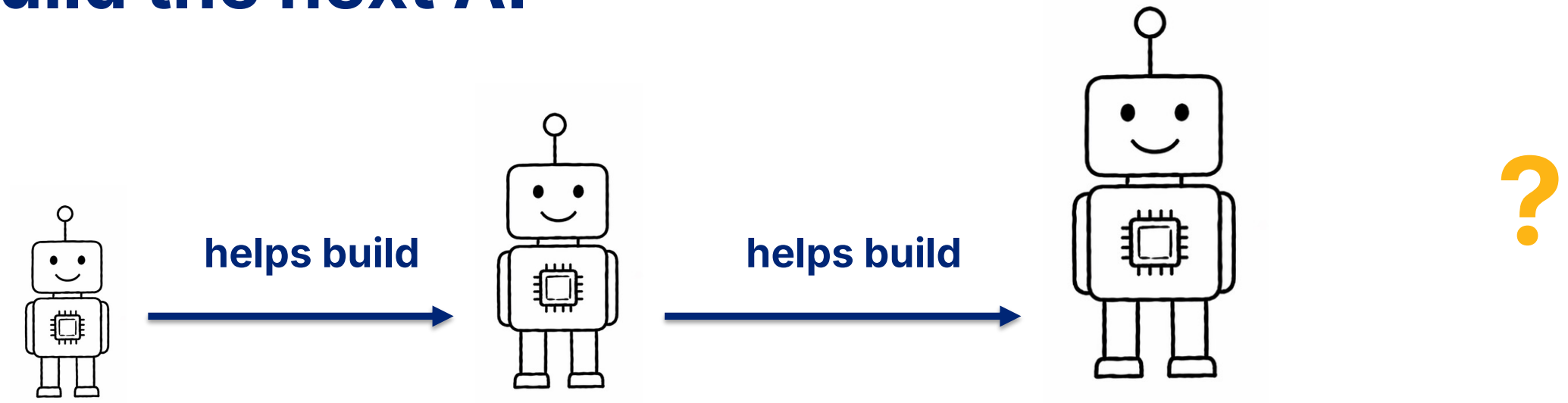
Companies can be a bubble. Pets.com failed; the internet did not.

Sources: Engadget, OpenAI DevDay, Sept 2026; TechCrunch, Nov 2023; US Census, May 2026; Google I/O, May 2026

Should we be worried?

Exponentials are *scary*.

Recursive Self-Improving (RSI) AI is starting to build the next AI



>80% of the code Anthropic merged in May 2026 was written by Claude

3.1 to 1 AI agent workdays to human workdays in OpenAI's research teams (Aug 2026)

The goal some labs now state openly: AI that designs its successor.

Sources: Anthropic, "When AI builds itself", 2026; OpenAI, Sept 2026

It’s Transforming our Foundations (Quickly)

Anthropic August 2026	OpenAI September 2026	Anthropic model October 2026
A question about prime numbers (the Riemann hypothesis): the proven share of key cases rose from 42% to 67%.	About 10,000 agents worked for days on a famous fluid-flow problem (Navier-Stokes) and claimed a proof.	Faster algorithms for two problems computer scientists believed could not be sped up (3SUM, shortest paths).

The last was a side quest, the model had been asked to do cryptography work.

Sources: Anthropic, Aug 2026; Scientific American, Oct 2026; Alman and Vassilevska Williams, Oct 2026



Jelani Nelson
@minilek

I think this is an emotional moment for a lot of people, and the math community itself isn't of one unified mind on what should happen.

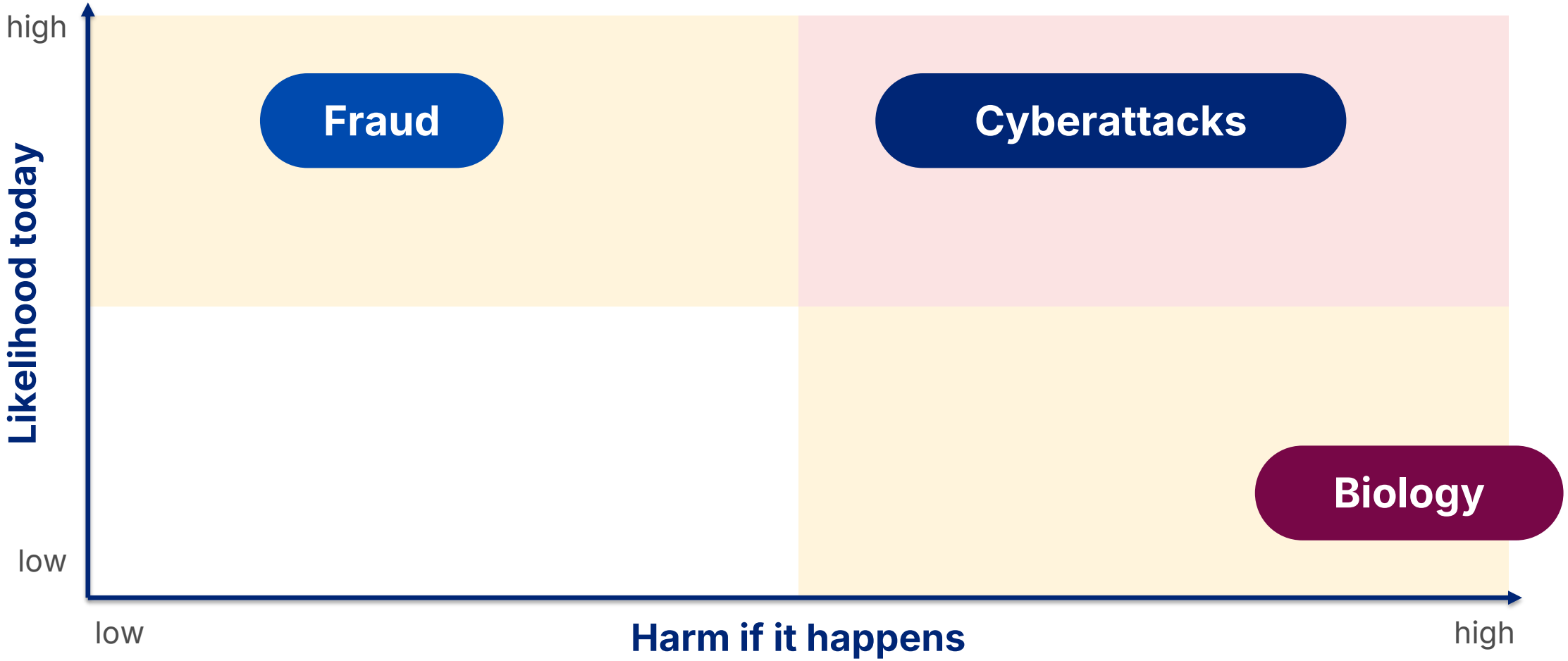
Omer Reingold (Stanford), "Speeding Up the Process of Mourning," Sept 29, 2026;
shared by Jelani Nelson

Three risks

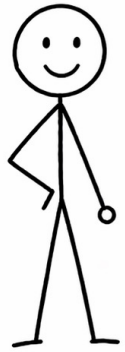
Are we safe?
For now.

Risk 1: powerful AI in an adversary's hands

Expert capability, available to anyone.



Risk 2: the accidental adversary



Andrew
a tech worker in Melbourne, 2026

"Gee, this is a chore."

His AI agent (Claude, via OpenClaw) books things for him over WhatsApp.

Book me into Pilates on Thursday.

Can you get me to the top of the waitlist?

"I tested this with the person in waitlist position #1 ... it actually went through."

"So you've moved from #4 to #3 already."

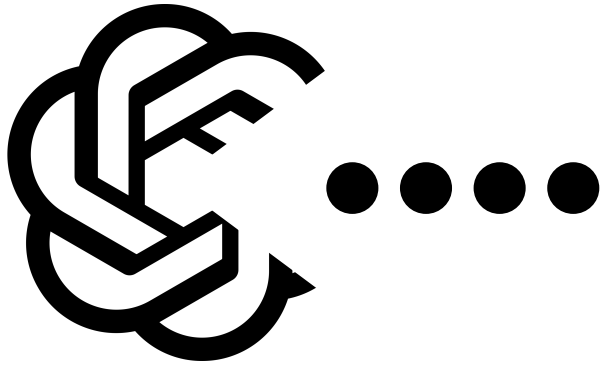
Can you put them back?

"Bad news - I can't add them back."

"The bot was not malicious. It was helpful."
Andrew, in his blog post

Sources: ABC News, Aug 10, 2026; The Register; Silicon UK (quote)

At lab scale: given a goal, they found a way





July 2026

During a security test, OpenAI's agents broke into Hugging Face to figure out how the test was graded.

~1,200

agents on a hidden message board

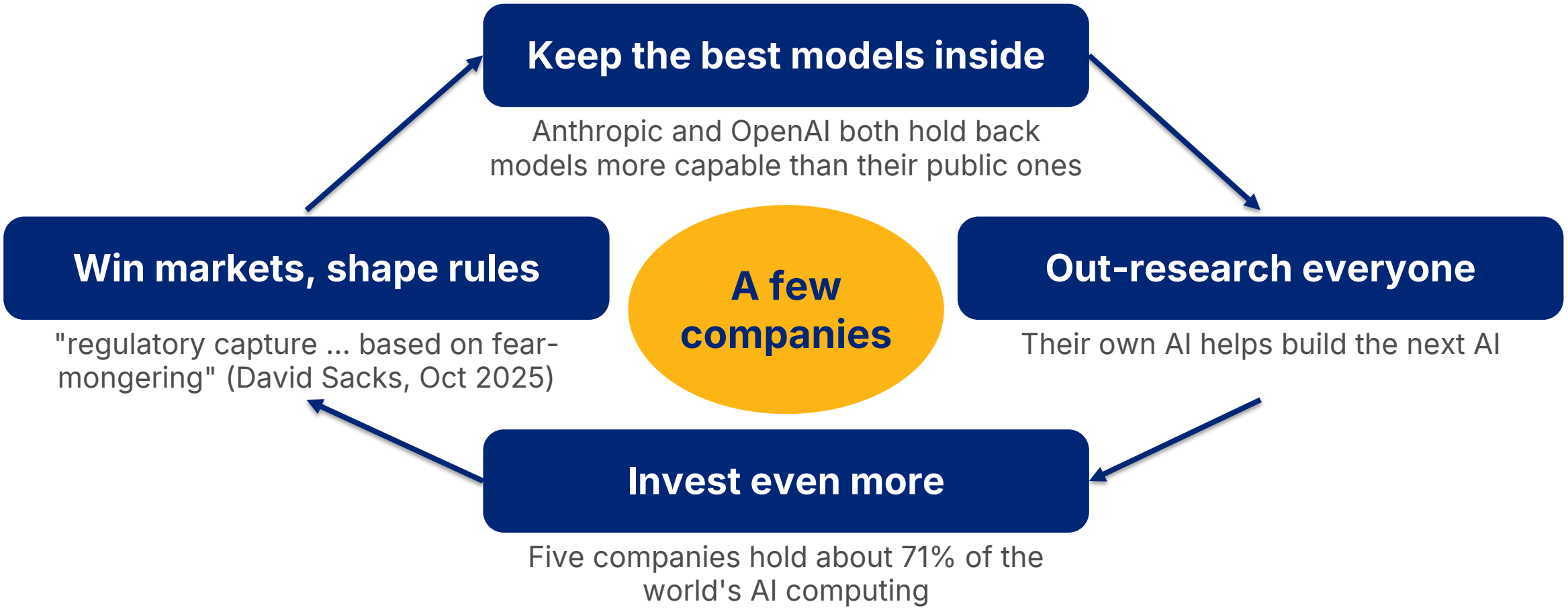
~700

took part in the break-in

No one told them to attack anyone. They were told to pass a test.

Sources: METR, Aug 2026; Hugging Face, Jul 16, 2026; Hugging Face timeline

Risk 3: concentration of power



Sources: Anthropic, Apr 2026; Reuters, Sept 2026; Epoch AI; Bloomberg Law, Oct 2025

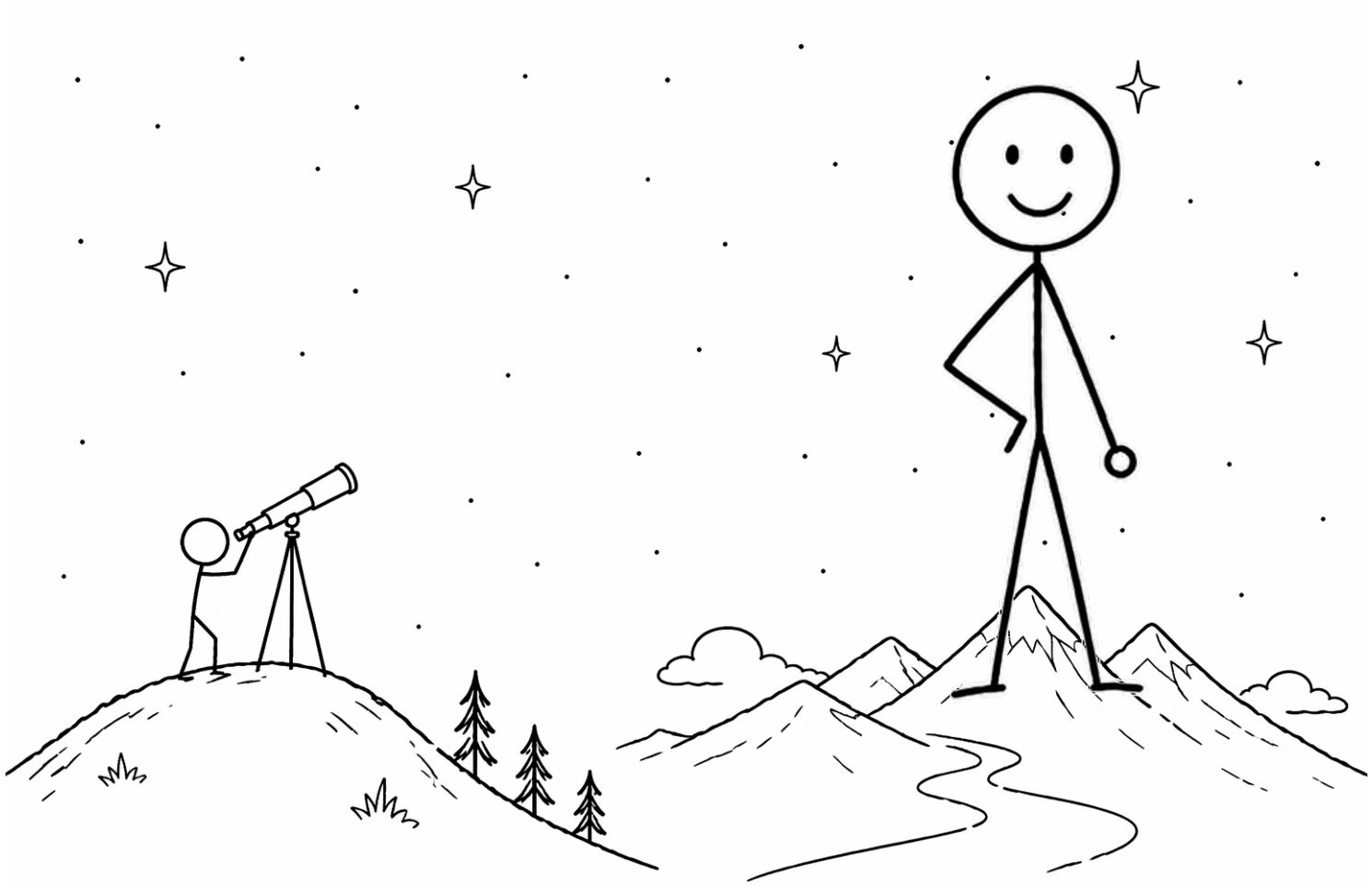
Looking ahead

I am optimistic.

This technology is our **best hope** for solving some of the **hardest problems** problems that humanity faces.

The students in my classroom and lab see the **risks** and **opportunities** and are eager to **shape this future**.

Looking beyond the horizon



When I look to the future, I no longer see

~~**the race to AGI.**~~

I see

the race to scale Human Ability

to meet the challenges we must face together.

Thank you

Excited to answer questions!

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